

Bhagy Tholiya

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Objective

Final-year CSE student at IITRAM (2023–27) with hands-on experience developing and deploying machine learning algorithms, deep learning models, and NLP-powered AI solutions. Proficient in Python, PyTorch, TensorFlow, FastAPI, and Docker. Eager to join Cloud Back as a Machine Learning Intern to build intelligent AI solutions that optimise cloud services performance and infrastructure.

Education

IITRAM – Institute of Infrastructure Technology Research and Management

Bachelor of Technology in Computer Science Engineering | CGPA: 7.54

Aug 2023 – Present

Ahmedabad, Gujarat

- Coursework: Machine Learning, Deep Learning, Natural Language Processing, Distributed Systems, Network Programming, Cryptography, DBMS

Projects

Real-Time Credit Card Fraud Detection — ML & Predictive Analytics Pipeline

[GitHub](#)

- Developed and implemented LightGBM + XGBoost machine learning algorithms for predictive analytics on **284,807 transactions**; achieved **ROC-AUC 0.9807** and sub-5ms inference via FastAPI REST API.
- Analysed and interpreted complex data to extract fraud pattern insights; built and trained models with SMOTE for pattern recognition; presented findings via MLflow dashboards.
- Deployed containerised AI solution using Docker Compose; Kafka streaming (2 tx/sec), EvidentlyAI drift monitoring, and Airflow DAG for automated retraining — optimised for scalable cloud infrastructure.

Document Layout Analysis & Intelligent OCR — Deep Learning + NLP System

[GitHub](#)

- Built and trained a deep learning Mask R-CNN (ResNet-50 FPN) model on **85K+ PubLayNet images** for layout pattern recognition; fine-tuned over 12,000 iterations across 5 layout classes.
- Developed NLP-powered pipeline integrating LangChain, FAISS, RAG architecture, and Anthropic Claude API for LLM-based summarisation; reduced manual review time by an estimated **40%**.
- Integrated AI-powered solution into existing document processing systems; delivered data-driven recommendations to stakeholders via structured JSON outputs and intelligence reports.

Chicken Disease Classification — Deep Learning for Predictive Analytics

[GitHub](#)

- Built and trained VGG16 deep learning transfer learning model achieving **94.83% accuracy** on image classification; reproducible DVC pipeline optimised for performance and scalability.
- Integrated model into Flask REST API; presented findings and recommendations via structured evaluation reports with production-ready logging and endpoint validation.

Global Economic Intelligence Dashboard — Data Analytics & AI Insights

[GitHub](#)

- Analysed World Bank dataset (**195 countries**, 5 macro indicators); applied K-Means clustering and statistical analysis to extract valuable insights and identify economic trends and patterns.
- Presented data-driven findings to stakeholders via Plotly choropleth maps, time-series charts, and Streamlit dashboard; delivered actionable comparative analysis reports.

Skills

ML / AI	Machine Learning Algorithms, Deep Learning, NLP, Predictive Analytics, Pattern Recognition, Computer Vision, Transfer Learning, SMOTE
Frameworks	Python, PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, OpenCV, Hugging Face, LangChain, RAG, FAISS, Pandas, NumPy
Data Science	Data Analytics, EDA, Feature Engineering, Statistical Analysis, Hypothesis Testing, Business Intelligence, Data Visualisation
Visualisation	Matplotlib, Seaborn, Plotly, Streamlit, Power BI
MLOps & Cloud	Docker, Kafka, Airflow, FastAPI, Flask, MLflow, DVC, CI/CD, AWS Basics, Git, GitHub, SQL, MySQL, PostgreSQL, MongoDB

Leadership & Activities

INFRA CYBERWISE-2026 — Infrastructure Cybersecurity Workshop

Jan 2026

Event Organiser, Cyber Club IITRAM

IITRAM, Ahmedabad

- Collaborated with cross-functional teams to organise two-day technical workshop; presented findings to **100+ attendees**; managed volunteers and end-to-end logistics.

CYBERWISE — Cybersecurity Workshop (Sponsored: CAWACH Kendra, Govt. of Gujarat)

Nov 2025

Event Organiser, Cyber Club IITRAM

IITRAM, Ahmedabad

- Co-organised two-day workshop on cyber law and fundamentals; managed **150+ registrations**; received award certificate for outstanding contribution.